Bambu Lab X1 Carbon Guide

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Important Policies

1. You may use a **maximum of two 3D printers at once.**
2. If you are printing in multiple colors at once or in any materials other than PLA, please find an associate to assist you.

Introduction

Bambu Lab **X1-Carbon** (referred to as “the Bambu”)

➢ The Bambu Lab X1-Carbon is the most popular fused deposition modeling (FDM) 3D printer in the Makerspace. They are available for free use to all Maryville College students. These printers have several advanced features, such

as LIDAR bed leveling and flow calibration, AI failure detection, and acceleration compensation, which allow them to print **extremely fast**. For these features to work, these printers require a short calibration time (approximately 5 to 10 minutes) before every print. **Please be patient while waiting for your print to begin.**

- ➢ Users are allowed to change filaments while the printer is not in use. To change filaments, remove the previous spool and insert the new one into the AMS. Speciality filaments (e.g. ABS, PETG, or TPU) must be requested, loaded, and used with the assistance of an associate. **These materials have different temperature tolerances and pose a greater risk of getting burned if not handled with care.**
- ➢ If you would like to print with multiple colors, ask an associate or officer before slicing your print.
- ➢ Please use Bambu Studio when slicing models for the Bambus, and download any sliced files onto the corresponding SD card.

Safety Tips

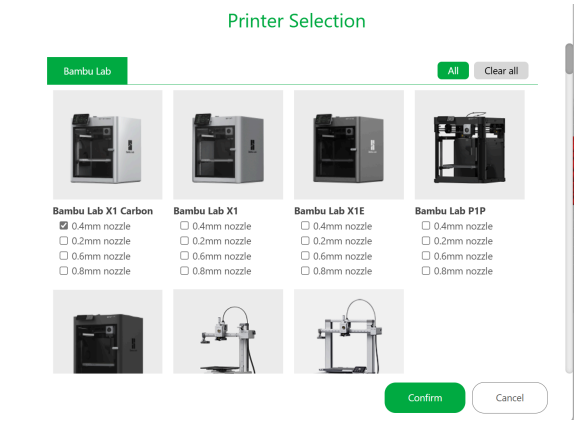
1. Heat
    - The nozzle of a Bambu will reach temperatures of 200+ °C during prints. **Touching this will burn you!**
    - In general, the print bed operates at lower temperatures (~60 - 70 °C) than the nozzle, but can still be hot to the touch. If possible, avoid touching the print bed during a print.
2. Movement
    - The toolhead (the housing for the nozzle) can move unexpectedly during a print, so be cautious of putting your hands inside the printer while it is active.
    - The 3D printer has a lot of mechanical movement from the X, Y, and Z axes. Do not touch the motor rails during a print, as you may get pinched!
3. Print Removal
    - Removing lines of filament from the print bed with your fingers can result in the filament sliding deep under your fingernail. Use a scraper instead.

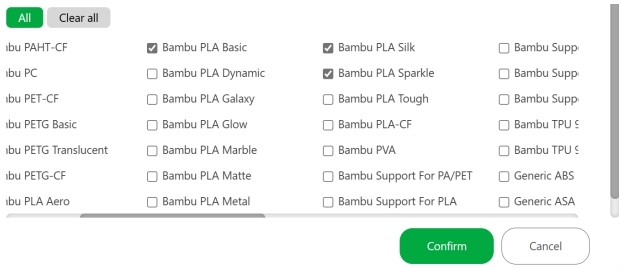
    - The scrapers are sharp. Using a scraper can result in sudden movement. Orient sharp tools away from yourself.
    - Supports can be brittle, and small pieces may break off and fly towards your eyes when removing them. Use safety glasses if necessary.
4. Errors/Suspicious Noises
    - Some error messages or warnings are due to dirty components on the Bambus and are still safe to continue printing.
    - However, always ask an associate, officer, returning member, or the MC3D advisor if you are unsure.
    - Please immediately stop printing if you hear the Bambus making suspicious noises, including but not limited to scraping, crashing, melting, crackling, or metallic thuds.

Bambu Studio

Setup

Follow the instructions in the Bambu Studio Quick Guide (https://wiki.bambulab.com/en/software/bambu-studio/studio-quick-start). Printer and filament types can be adjusted at any time. Be sure to check the 0.4 mm Nozzle under the Bambu Labs X1 Carbon.





Slicing Settings

There are many different settings that can be adjusted inside the Bambu Studio slicer. However, we will cover the most common and crucial settings here.

a. **Nozzle Diameter (mm):** Our printers only use 0.4 mm nozzles.
b. **Filament:** Bambu-branded filaments, including PLA Matte, PLA Basic, PLA Sparkle, and PLA Metal, are stocked depending on members' needs. Off-brand filaments and filaments with higher temperature tolerances may also be used. Ask an associate for details.
c. **Layer Height:** This indicates how large the gaps are between each layer. The lower the layer height, the smoother (higher quality) the print will look. The default value is 0.2 mm. Lower values are suggested for aesthetic prints but will increase print time.
d. **Wall Loops:** The number of solid loops that the printer will create on the outside of your part. More loops mean longer print times, but will produce a slightly stronger part.
e. **Infill Density (%):** This indicates how dense your print is. The default value is 15%. 25% is recommended for sturdier parts and 40%-60% for weight-bearing parts.

f. **Sparse Infill Pattern:** Rectilinear and grid are good for typical prints and prototyping. Gyroid is the overall recommendation for its structural integrity.
g. **Supports:** A warning message might appear if you slice a model with floating parts without supports. Supports are extra material that prevent the nozzle from printing on air and connect floating regions to the rest of the print. Tree supports on the bed plate are recommended.
h. **Rafts, Brims, & Skirts:** All different adhesion types to help a print stick to the bed. Unless your print might have an issue adhering to the bed, a skirt should suffice.
    i. Rafts are several layers under the print to help it stick to the bed. Rafts are useful for larger models or models with little surface area touching the plate.
    ii. Brims are lesser versions of rafts that trace the inside or outside of the print.
    iii. Skirts act as pre-extrusion.

Troubleshooting

❌Filament Ran Out

Fix: Remove the empty spool from the AMS and replace it with a new one. Enter the filament into the AMS. If the problem persists or the spool in question is not empty, then a piece of filament may be stuck in the tubing or extruder.

🦞Failed to Pull Back Filament From Toolhead

Fix: This one causes headaches. First, remove the spool from the AMS slot. Check the tubing, AMS hub, and toolhead for any signs of broken filament strands. If no luck, try running filament through the extruder. If that doesn't work, open up the AMS and check the tubing for filament. If the issue happens to not be caused by broken filament, consider changing the specific AMS slot motor.

🛢️Z-axis Rods/Screws Need Cleaning

Fix: Gently pat the Z-axis rods with a paper towel and isopropyl alcohol. Then, apply a little lubricating grease to each one.

🪂Chute is Full

Fix: The printer is experiencing indigestion. Help it by clearing out the poop bucket behind it into a Ziploc bag. Place that bag into one of the recycling bins in the Makerspace.

FAQ

Q: I want to print in a specific color that is not loaded. Can I load the color I want?

A: Yes, you may use any PLA filaments that are out for general use. Pull out the filament you intend to replace, and insert the new filament. The AMS should automatically feed in the new color. Check that the filament's info is correct in the Bambu.

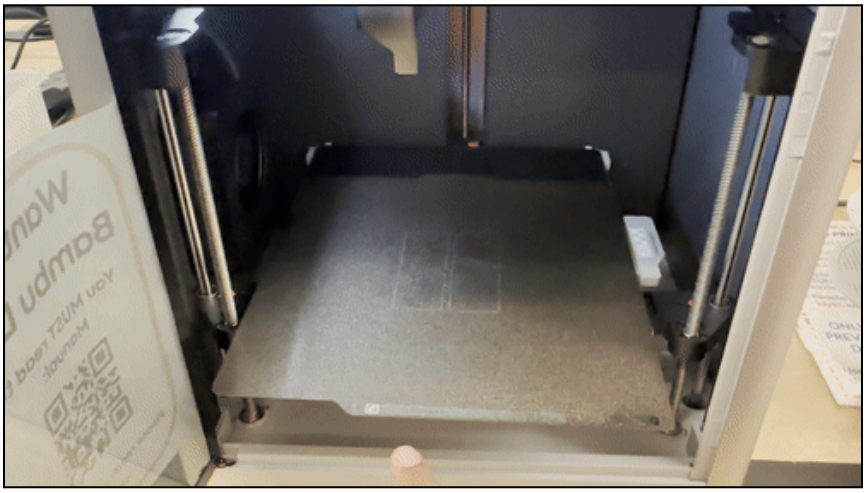
Q: The printer has four spools of filament at once. Can I print different colors in the same print?
A: This is up to the discretion of the associate on duty since multicolor prints produce an incredible amount of waste by changing filaments with every layer.

Q: My print failed and looks like a pile of spaghetti. Is something wrong with the printer?
A:

Spaghetti failures happen for several different reasons. Some prints have adhesion issues with the heatbed. Sometimes, the first few layers don't stick. Try adding glue to the build plate. If the nozzle keeps knocking the print off, try adding a raft or brim to the print. This extra piece helps the print stay on the bed while it is printing.

If the print has supports, try adjusting the support settings.

Q: The printer says the bed is installed improperly, or that z-level calibration failed, or that something about leveling bed forces. What's wrong?
A: Make sure the bed is installed correctly. Make sure it is facing the right direction and isn't stuck on the plastic nubs on the back of the printer. See the video below for visual instructions.



Q: The printer says my model is sliced for the wrong printer or plate? What's the difference between bedplates?
A:
Texture Plate: A dual-sided, versatile plate for a durable, scratch-resistant surface to print different engineering-grade materials
Cool Plate: A smooth surface plate ideal for PLA, offering reliable adhesion and easy removal at lower temperatures

Q: A red light is appearing on the AMS, and the printer is saying that the AMS is overloaded.
A: This could be for a couple of reasons. First, try reloading the filament. The filament may be caught in the AMS motor. If the issue persists, filament may be stuck in the tubing, or the motor may need to be replaced. Follow this video for assistance in taking apart the AMS.

Q: The printer ran out of filament/has an error mid-print! What should I do?
A: If a spool of filament runs out mid-print, replace the empty spool with the same or a similar color. If an error occurs (e.g. spaghetti failure), stop the print and try again with adjusted settings.

Q: I'm printing multiple parts on the same plate, and one of them got messed up. What can I do?
A: The Bambu printers actually have a convenient feature for "skipping" over certain objects mid-print without needing to stop the print entirely. You can read more in the skipping objects page on Bambu's help website.

Q: How long does it take to print?
A: Print time depends on a lot of variables and varies with every print. Larger, denser, and finer-quality models will take longer.